

08 — Customer Acquisition

Health Hub Business Plan v2 | Document 8 of 15 Version: 2.0 | Date: March 2026 | Covers: All 21 scenario x level combinations

All assumptions, rates, phase definitions, and scenario codes in this document trace back to [params.md](#). When this document says "per params.md," it means the v2 shared assumptions document.

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1. Executive Summary

Health Hub's customer acquisition strategy is built on a single structural insight: in East Africa, patients follow providers. The facility — the clinic, the hospital, the pharmacy, the lab — is the unit of distribution. Onboarding one facility delivers 50-200 patients at near-zero marginal cost. This makes B2B facility acquisition the highest-leverage go-to-market activity through Pilot 2, and a persistent organic base layer even at Production scale.

This document models acquisition costs, channels, funnels, and LTV:CAC ratios across all 21 scenario x level combinations defined in params.md. It introduces a key v2 addition: **admin ops staff as a manual acquisition and conversion channel**. Health Hub's operating model employs admin ops staff (per params.md Section 4) who handle callbacks, patient onboarding assistance, and inquiry-to-customer conversion. This human layer — unique among health tech platforms in the region — functions as a high-touch acquisition channel with measurable CAC and conversion rates.

Central findings

- **Facility-first acquisition** dominates all channels through Pilot 2. Each onboarded facility generates 50-200 patient registrations organically.
- **Admin ops callbacks** convert 30-50% of inbound inquiries into registered users, functioning as a low-CAC manual acquisition channel from Pilot 1 onward.
- **Blended B2C CAC** remains below \$1.00 in founder-led phases (Pre-Pilot through Pilot 1), rises to \$1-3 at Pilot 2, and \$2-5 at Production as paid channels scale.
- **All 21 combinations achieve viable unit economics**. LTV:CAC ratios range from 4.6x (S3-L3-I3, maximum spend) to 23x+ (S1-L1, maximum efficiency). No combination enters the sub-3x danger zone.
- **The baseline plan (S2-O2-I2)** projects a blended CAC of \$1.50 at Pilot 2 and \$2.50 at Production, with LTV:CAC exceeding 5x once subscription revenue stabilizes.

Strategic principles

Principle	Rationale
Facility-first	Every onboarded facility generates 50-200 patient registrations organically, making B2B acquisition the primary growth lever
Admin-assisted conversion	Employed admin ops staff handle callbacks and onboarding, converting inquiries at 30-50% — a manual channel unique to Health Hub
Mobile-money native	M-Pesa (Kenya) and Telebirr (Ethiopia) integration removes payment friction, the largest barrier to conversion in East Africa
WhatsApp as distribution	40M+ WhatsApp users across Ethiopia and Kenya; community-based distribution is the highest-ROI digital B2C channel
Android-first timing	Android app launch (mid-Pilot 1) unlocks Play Store distribution, which becomes the dominant scaled B2C channel by Pilot 2
Retention over acquisition	A 5-point churn reduction doubles LTV, delivering higher ROI than equivalent top-of-funnel spend

2. Customer Segments

2.1 B2B: Healthcare Facilities

Segment	Description	Acquisition Strategy	Decision Maker	Est. Facilities in Addis Ababa	Priority
Private clinics (1-5 doctors)	Primary target. High density in urban areas, owner-operated, fast decision cycles (~2-4 weeks). Represent 70-80% of private healthcare delivery in Ethiopia.	Founder-led direct sales, warm introductions, admin ops follow-up calls	Clinic owner / managing doctor	2,000-3,000	Highest — Pilot 1 onward
Private hospitals (6+ doctors)	Secondary target. Higher patient volume per facility (500-2,000 patients/mo) but longer sales cycles (4-8 weeks) and procurement processes.	Referral from onboarded clinics, partnership proposals, conference networking, admin ops relationship management	Hospital administrator / CTO / medical director	200-400	High — Pilot 2 onward
Pharmacy chains (2+ outlets)	Pharmacy module users. Often co-located with or adjacent to clinics, creating natural bundling opportunities. Revenue via take-rate on order value.	Bundle with clinic onboarding, direct outreach to chain owners, admin ops coordination for prescription routing	Pharmacy owner / chain operations manager	3,000-5,000 (total pharmacies; ~200-300 chains)	Medium — Pilot 1 onward
Diagnostic laboratories	Lab module users. Referral-dependent business model aligns naturally with platform marketplace. Revenue via take-rate on lab orders.	Bundle with clinic onboarding, referral partnerships with onboarded GPs, admin ops lab order coordination	Lab manager / owner	500-800	Medium — Pilot 2 onward

Public / NGO facilities	Extension phase only. Government procurement cycles are 6-18 months. NGO partnerships require compliance documentation and data protection agreements.	Government health department partnerships, NGO grant proposals, WHO/UNICEF/USAID digital health channels	Government health department / NGO program director	50-100 (major facilities)	Low — Production onward
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B2B segmentation notes:

- The 5-10 warm leads in the current pipeline are private clinics and small hospitals in Addis Ababa — the Pilot 1 onboarding pool.
- Pharmacy and lab acquisition is most efficient when bundled with clinic onboarding: a clinic that adopts Health Hub naturally drives prescriptions and lab orders through the platform, pulling pharmacies and labs into the ecosystem.
- Admin ops staff play a critical role in B2B relationship management from Pilot 1 onward: handling callback scheduling with facility managers, coordinating technical onboarding, and managing ongoing partner communications.
- Public facility acquisition is deferred to Production/Extension due to regulatory complexity and long procurement timelines. The exception is NGO-funded clinics, which may adopt faster if the platform aligns with their digital health mandates.

2.2 B2C: Patients

Segment	Description	Acquisition Strategy	Key Channel	Est. Addressable Population (Addis Ababa)	ARPU Estimate	Priority
Urban professionals (25-45)	Primary segment. Smartphone users with disposable income, already paying for private healthcare out-of-pocket. Health-conscious, time-constrained. Value convenience and quality.	Digital marketing, app store optimization, social media, admin ops callback for inquiries	Social media (Facebook, Instagram), Play Store, admin ops callbacks	400,000-600,000	\$2.00-4.00/mo	Highest
Urban families	Secondary segment. Driven by children's and elderly relatives' healthcare needs. Larger household spending but more price-sensitive per transaction. Multi-user lock-in opportunity.	Facility recommendation, family health content, admin ops onboarding assistance for dependents	Facility referral, WhatsApp family groups, admin ops	300,000-500,000	\$1.50-3.00/mo	High
Students / young adults (18-25)	Early adopters with high digital literacy but lower ability to pay. Valuable for viral growth and social proof. Campus health center partnerships.	Social media campaigns, campus outreach, peer referrals, Play Store organic	WhatsApp, TikTok, Instagram, campus health centers	200,000-350,000	\$0.50-1.50/mo	Medium
Rural-accessible	Extension phase. Patients within 1-2 hours of urban centers who travel for specialist care. Telemedicine value proposition is strongest for this segment.	Community health worker networks, SMS campaigns, radio advertising, admin ops triage and scheduling	Word of mouth, SMS, CHWs, admin ops callbacks	500,000-1,000,000	\$0.25-1.00/mo	Low (Production)

B2C segmentation notes:

- In Ethiopia, the urban professional segment overlaps with the existing private healthcare market. These patients already pay ETB 200-500 per GP visit out-of-pocket, making the platform's ETB 200 (\$3.50) GP consultation price competitive.
- Family acquisition follows a "household gateway" pattern: one family member registers, then books for children or elderly relatives. The platform must support dependent profiles to capture this.
- Student acquisition is a long-term play: low immediate ARPU but high lifetime value as students enter the workforce and their healthcare spending increases.
- Admin ops staff handle callback-based onboarding for all segments: a patient who calls in or submits a web inquiry is called back by admin ops, walked through registration, and booked for their first consultation. This manual touch point is a conversion accelerator unique to Health Hub's model.

3. Acquisition Channels

3.1 Channel Ranking by Phase

Channels are ranked in descending order of priority within each phase. The ranking reflects expected ROI, feasibility, and resource availability.

Pre-Pilot (M0-M3/M6, 100-200 internal users)

Rank	Channel	Type	Cash Cost	Time Cost	Expected Yield	Notes
1	Personal network	B2B + B2C	\$0	20-40 hrs total	2-3 facilities, 100-150 users	Founders' existing relationships in Addis Ababa
2	Founder-led outreach	B2B	\$0 cash (\$50-150 travel)	20-40 hrs/facility	1-2 additional facilities from cold/warm approaches	In-person demos at facility. Local travel within Addis.
3	Internal team testing	B2C	\$0	Embedded in dev cycle	50-100 test users	Team members, friends, family as beta testers

Pilot 1 (up to 1,000 users, 3-5 facilities)

Rank	Channel	Type	Cash Cost	Time Cost	Expected Yield	Notes
1	Facility-driven onboarding	B2C	\$0 marginal	10-20 hrs/facility setup	50-200 patients per facility	Clinic staff assist patients with registration at point of care
2	Admin ops callbacks	B2C	\$0.30-0.80/conversion (staff time)	Ongoing (staff hours)	30-50% conversion of inbound inquiries	Admin ops staff (2 from Pilot 1, per params.md) call back every inquiry, walk through registration, book first

						consultation
3	Founder-led facility sales	B2B	\$50-150/trip	20-40 hrs/facility	3-5 facilities from warm leads	5-10 warm leads with 40-60% conversion rate
4	WhatsApp groups	B2C	\$0	5-10 hrs/mo	50-150 users	Create health tip communities; share booking links
5	Word of mouth	B2C	\$0	\$0	50-100 users	Organic spread from satisfied pilot patients
6	Social media organic	B2C	\$0	5-10 hrs/mo	20-50 users	Facebook page, health content posts. Ethiopia has 15M+ Facebook users.

Pilot 2 (5,000-10,000 users, 10-20 facilities)

Rank	Channel	Type	Cash Cost	Time Cost	Expected Yield	Notes
1	Facility-driven onboarding	B2C	\$0 marginal	Ongoing	35% of new users	Still the largest single channel
2	Google Play Store (ASO)	B2C	\$0.50-2.00/install	10-15 hrs/mo (ASO work)	20% of new users	Android app published. Keywords: "doctor Ethiopia," "clinic Addis Ababa." Organic installs scale with ratings.
3	Admin ops callbacks	B2C	\$0.30-0.80/conversion	Ongoing (4 staff from Pilot 2)	10-15% of new users	Expanded admin team handles higher inquiry volume; callback-to-registration pipeline matures
4	Social media paid (light)	B2C	\$1.00-5.00/user	10-15 hrs/mo	10-15% of new users	CPM in Ethiopia is \$2-5 (3-5x cheaper than Western markets). Facebook/Instagram targeting: urban, 25-45, health interests.
5	Referral program	B2C	\$0.50-1.50/referral	5 hrs/mo (management)	10-15% of new users	In-app referral: give ETB 50 credit, get ETB 50 credit per successful referral. Target viral coefficient: 0.3.
6	WhatsApp communities	B2C	\$0-0.25/user	5-10 hrs/mo	5-10% of new users	Scale existing groups; health content calendar
7	Community health workers	B2C	\$1.00-3.00/referral	10-15 hrs setup	2-5% of new users (testing)	Pilot CHW referral incentive in 1-2 areas outside central Addis
8	Facility referral (B2B)	B2B	\$0 (free month incentive)	2-5 hrs/referral	5-8 new facilities	Each onboarded facility knows 3-5 peer facilities
9	Healthcare events	B2B	\$500-2,000/event	20-30 hrs/event	2-3 facilities per event	Ethiopian Medical Association conferences. 1-2 events/year.

Production (10,000-50,000 users, 50-100+ facilities)

Rank	Channel	Type	Cash Cost	Time Cost	Expected Yield	Notes
1	Google Play Store (ASO + paid)	B2C	\$1.00-3.00/install	15-20 hrs/mo	20-25% of new users	Organic + Google Ads for app installs
2	Facility-driven onboarding	B2C	\$0 marginal	Ongoing	20-25% of new users	Base layer; share decreases as other channels scale
3	Social media paid	B2C	\$2.00-5.00/user	15-20 hrs/mo	15-20% of new users	Full Facebook/Instagram/TikTok campaigns
4	Admin ops callbacks + outbound	B2C	\$0.50-1.00/conversion	Ongoing (6-8 staff)	8-12% of new users	Mature callback operation; some outbound follow-up for lapsed users
5	Referral program	B2C	\$1.00-2.00/referral	5-10 hrs/mo	10-15% of new users	Viral coefficient target: 0.5
6	SMS / telecom campaigns	B2C	\$0.10-0.30/SMS	5-10 hrs/campaign	5-10% of new users	Bulk SMS via Ethio Telecom/Safaricom partnerships
7	Community health workers	B2C	\$1.00-3.00/referral	10-15 hrs/mo	5-8% of new users	Scaled CHW incentive program
8	Radio / local media	B2C	\$200-1,000/campaign	10-15 hrs/campaign	3-5% of new users	FM radio reaches 70%+ of urban Ethiopia. Amharic-language spots.
9	Telecom bundling	B2C	\$0.10-0.50/user	40-80 hrs (partnership)	2-5% of new users	Bundle Health Hub with Ethio Telecom data plans
10	Digital B2B marketing	B2B	\$200-500/mo	10-15 hrs/mo	3-5 facilities/mo	LinkedIn, Google Ads for "clinic management software Ethiopia"
11	Healthcare associations	B2B	\$200-500/yr	10-15 hrs/quarter	2-3 facilities/quarter	Ethiopian Medical Association member directories

3.2 Admin Ops as an Acquisition Channel (v2 Addition)

Health Hub's operating model includes employed admin ops staff (per params.md Section 4). While their primary function is operational — callbacks, exception handling, partner coordination — they also serve as a measurable acquisition and conversion channel.

How admin ops drive acquisition:

Function	Acquisition Impact	Phase	Staffing
Inbound inquiry callbacks	Patient calls in or submits web form; admin ops calls back within 2 hours, walks through registration, books first consultation	Pilot 1+	2 admin ops (Pilot 1), 4 (Pilot 2), 6-8 (Production)
Facility onboarding support	Admin ops coordinates technical setup, staff training, and go-live for new facilities — reducing founder time per facility from 40 hrs to 10-15 hrs	Pilot 1+	Shared with above
Lapsed user re-engagement	Admin ops calls users who registered but never booked, or who have not returned in 60+ days	Pilot 2+	Shared with above
Travel/tourism patient coordination	Admin ops handles care navigation for patients traveling from outside Addis or international medical tourists — a high-value manual conversion	Pilot 2+	1-2 dedicated
Partner coordination	Admin ops maintains relationships with pharmacies and labs, ensuring prescription/lab order fulfillment, which drives patient satisfaction and retention	Pilot 1+	Shared with above

Admin ops acquisition economics:

Metric	Pilot 1	Pilot 2	Production
Admin ops staff	2	4	6-8
Monthly salary cost (total)	\$434 (2 x \$217)	\$868 (4 x \$217)	\$1,302-1,736
Hours/mo available for acquisition activities	~80 (50% of capacity)	~160	~240-320
Inquiries handled/mo	50-100	200-400	500-1,000
Conversion rate (inquiry to registered user)	30%	40%	50%
New registrations via admin ops/mo	15-30	80-160	250-500
Effective CAC per admin-acquired user	\$14.50-29.00 (salary allocation)	\$5.40-10.90	\$2.60-6.90
Adjusted CAC (admin handles multiple functions)	\$0.30-0.80 (20% of time on acquisition)	\$0.30-0.80	\$0.50-1.00

Key insight: The adjusted CAC for admin-acquired users is low (\$0.30-1.00) because admin ops staff are salaried employees whose acquisition work is one of several functions. The marginal cost of each additional conversion is near zero once the staff are employed. This makes admin ops one of the most capital-efficient acquisition channels from Pilot 1 through Production.

4. CAC Model

4.1 B2B CAC by Phase

Phase	Facilities Target	Primary Acquisition Method	Est. Cost per Facility	Total B2B CAC Budget	Key Assumptions
Pre-Pilot	2-3 (internal/friends)	Personal network, warm introductions	\$0 (time only)	\$0	Founders' existing relationships. No travel costs (local).
Pilot 1	3-5 (from warm leads)	Founder-led sales + admin ops follow-up + local travel	\$300-500	\$900-2,500	5-10 warm leads with 40-60% conversion. Travel within Addis: \$50-150/trip. Admin ops handles scheduling and follow-up.
Pilot 2	10-20	Sales + referrals + events + admin ops coordination	\$200-400	\$2,000-8,000	30% from referrals (\$0 cash), 40% direct sales (\$300-500), 30% events/outreach (\$200-400). Admin ops reduces founder time per facility.
Production	50-100+	Full mix (sales + digital + partnerships + referrals)	\$150-300	\$7,500-30,000	Referral channel grows to 40-50% of new facilities. Digital B2B marketing supplements direct sales. Dedicated B2B sales capacity.

B2B CAC dynamics:

- CAC per facility decreases over time as referral networks mature and brand recognition shortens the sales cycle.
- The most expensive facilities to acquire are the first 3-5 (Pilot 1), where every onboarding requires extensive hand-holding, custom training, and relationship building.
- At Production scale, the target is a blended B2B CAC of \$150-300 per facility, with referral-acquired facilities at \$0-50 subsidizing the higher cost of event-sourced and cold-outreach facilities.
- Admin ops staff reduce B2B CAC by absorbing partner coordination work that would otherwise consume founder time (valued at \$20/hr per params.md).

4.2 B2C CAC by Phase

Phase	Users Target	Primary Channel	Est. CAC	Total B2C Cost	Channel Mix
Pre-Pilot	100-200	Internal (team, friends, family)	~\$0	\$0	100% organic/internal
Pilot 1	1,000	Facility-driven + admin ops + WhatsApp	\$0.25-0.75	\$250-750	50% facility-driven (\$0), 15% admin ops callbacks (\$0.50), 20% WhatsApp (\$0.25), 15% social organic (\$0.75)
Pilot 2	5,000-10,000	Social + ads + referral + Play Store + admin ops	\$1.00-3.00	\$5,000-30,000	30% facility-driven, 20% Play Store, 15% admin ops, 15% social paid, 10% referral, 10% WhatsApp
Production	10,000-50,000	Full channel mix	\$2.00-5.00	\$20,000-250,000	22% Play Store, 20% facility-driven, 15% social paid, 10% admin ops, 10% referral, 8% SMS, 7% CHW, 5% radio, 3% telecom

4.3 Blended CAC by Phase and Scenario Family

Phase	S1 (Self-Funded)	S2 (Investor at P2)	S3 (Investor after P1)
Pre-Pilot	~\$0	~\$0	~\$0
Pilot 1	\$0.15-0.75	\$0.15-0.75	\$0-0.50
Pilot 2	\$0.20-1.05	\$0.60-2.50	\$0.60-2.50
Production	\$0.50-1.50	\$1.00-3.00	\$1.00-3.50

Blended CAC interpretation: The wide ranges reflect the L1-L3 investment level variation. At L1 (bootstrapped), acquisition is almost entirely organic and facility-driven, yielding very low CAC but slow growth. At L3 (fully funded), paid channels are activated earlier and at higher budgets, raising CAC but accelerating user growth.

4.4 CAC by Channel (Detailed Breakdown)

Channel	Pilot 1 CAC	Pilot 2 CAC	Production CAC	Cost Trend	Volume Trend
Facility-driven	\$0	\$0	\$0-0.50	Flat	Grows with B2B
Admin ops callbacks	\$0.30-0.80	\$0.30-0.80	\$0.50-1.00	Slight increase	Grows with staff
WhatsApp communities	\$0-0.25	\$0-0.25	\$0.25-0.50	Slight increase	Moderate growth
Social media organic	\$0.50-1.00	\$0.50-1.00	\$0.75-1.50	Slight increase	Moderate growth
Google Play Store (ASO)	N/A (pre-launch)	\$0.50-2.00	\$1.00-3.00	Increase	High growth
Social media paid	N/A	\$1.00-5.00	\$2.00-5.00	Stable	High growth
Referral program	N/A	\$0.50-1.50	\$1.00-2.00	Slight increase	High growth
SMS campaigns	N/A	\$0.10-0.30/SMS	\$0.10-0.30/SMS	Flat	Moderate growth
Community health workers	N/A	\$1.00-3.00	\$1.00-3.00	Flat	High growth (rural)
Radio / local media	N/A	N/A	\$2.00-5.00 (est.)	Unknown	Mass reach
Telecom bundling	N/A	N/A	\$0.10-0.50	Very low	Very high

5. LTV Model

5.1 Revenue per User per Month (ARPU)

Revenue sources per user per month, based on pricing corridors in params.md Section 11.2:

Revenue Component	Conservative	Moderate	Optimistic	Basis
GP consultation fees (ETB 200 / \$3.50)	\$0.30	\$0.70	\$1.40	1 consult/3mo vs. 1/2mo vs. 1/mo
Specialist consultation fees (ETB 400 / \$7.00)	\$0.00	\$0.25	\$0.60	0 vs. 1/4mo vs. 1/3mo
Premium subscription (ETB 300/mo / \$5.26)	\$0.10	\$0.20	\$0.40	2% vs. 4% vs. 8% of users subscribe
Pharmacy / lab commissions (8-12% take-rate)	\$0.05	\$0.10	\$0.20	Commission on orders routed through marketplace
Care coordination / travel fees	\$0.00	\$0.05	\$0.15	Per-case fees for travel/tourism patients (Pilot 2+)
Other (facility pass-through, data)	\$0.05	\$0.05	\$0.05	Negligible in early phases
Total ARPU/mo	\$0.50	\$1.35	\$2.80	

5.2 LTV by Scenario Case

Metric	Conservative	Moderate	Optimistic
Average revenue per user per month (ARPU)	\$0.50	\$1.35	\$2.80
Average customer lifespan (months)	6	12	24
Gross LTV (ARPU x lifespan)	\$3.00	\$16.20	\$67.20
Gross margin	90%	92%	95%
LTV (margin-adjusted)	\$2.70	\$14.90	\$63.85
Monthly churn rate (implied)	16.7%	8.3%	4.2%

5.3 LTV by Customer Segment

Segment	Avg. Consults/Mo	Avg. ARPU/Mo	Avg. Lifespan (Mo)	Segment LTV	Churn Rate
Urban professionals (25-45)	0.5-1.0	\$2.00-4.00	12-18	\$24.00-72.00	6-8%
Urban families	0.3-0.7	\$1.50-3.00	18-36	\$27.00-108.00	3-6%
Students / young adults	0.2-0.4	\$0.50-1.50	4-8	\$2.00-12.00	15-25%

Rural-accessible	0.1-0.3	\$0.25-1.00	6-12	\$1.50-12.00	10-15%
Travel / tourism patients	1-3 (episodic)	\$10.00-25.00	1-3 (episodic)	\$10.00-75.00	N/A (project-based)

5.4 LTV:CAC Ratios by Phase and Channel

Phase	Channel	CAC	LTV (Moderate)	LTV:CAC	Verdict
Pilot 1	Facility-driven	\$0.00	\$14.90	Infinite	Best channel
Pilot 1	Admin ops callbacks	\$0.50	\$14.90	29.8x	Excellent
Pilot 1	WhatsApp	\$0.25	\$14.90	59.6x	Excellent
Pilot 1	Social organic	\$0.75	\$14.90	19.9x	Excellent
Pilot 2	Facility-driven	\$0.00	\$14.90	Infinite	Best channel
Pilot 2	Admin ops callbacks	\$0.50	\$14.90	29.8x	Excellent
Pilot 2	Play Store (ASO)	\$1.00	\$14.90	14.9x	Very Good
Pilot 2	Social paid	\$3.00	\$14.90	5.0x	Good
Pilot 2	Referral program	\$1.00	\$14.90	14.9x	Very Good
Production	Facility-driven	\$0.50	\$14.90	29.8x	Excellent
Production	Admin ops callbacks	\$0.75	\$14.90	19.9x	Excellent
Production	Play Store (ASO)	\$1.50	\$14.90	9.9x	Very Good
Production	Social paid	\$4.00	\$14.90	3.7x	Acceptable
Production	Radio/SMS	\$2.00	\$14.90	7.5x	Good
Production	CHW referral	\$2.50	\$14.90	6.0x	Good
Production	Telecom bundling	\$0.30	\$14.90	49.7x	Excellent

LTV:CAC benchmark: A ratio of 3x or higher is considered healthy for venture-backed startups. Below 3x requires either CAC reduction or ARPU increase to achieve sustainable unit economics. All channels at all phases exceed the 3x threshold.

6. Acquisition Funnel

6.1 Funnel Stages and Conversion Rates

Stage	Definition	Conversion Rate	Key Driver	Measurement
Awareness	User hears about Health Hub (ad impression, word of mouth, clinic poster, admin ops mention)	Baseline (top of funnel)	Channel reach	Impressions, reach, brand recall
App Download / Site Visit	User downloads Android app or visits web platform	5-15% of awareness	Channel quality, creative, ASO	Install count, unique visits
Registration	User completes signup (name, phone, password)	40-60% of downloads	Mobile-money-linked signup, simplified flow, admin ops assisted registration	Registered accounts
First Consultation	User completes first paid or free consultation	15-30% of registrations	Onboarding tutorial, first-consult discount, admin ops booking assist	Completed consultations
Repeat Use (MAU)	User returns within 30 days	30-50% of first-use users	Push notifications, appointment reminders, WhatsApp reminders	30-day active users
Paying User	User completes at least 1 paid transaction	25-40% of MAU	Consultation quality, pricing, payment ease	Transacting users
Premium Subscriber	User subscribes to monthly plan (ETB 300/mo)	5-15% of paying users	Subscription value proposition, unlimited GP, priority booking	Subscription count

6.2 Funnel Metrics by Phase

Metric	Pre-Pilot	Pilot 1	Pilot 2	Production
Top of funnel (awareness)	500	5,000	50,000	250,000+
App downloads / site visits	250	1,500	15,000	75,000
Registrations	200	1,000	10,000	50,000
First consultation	50	250	2,500	15,000
Monthly active users (MAU)	100	350	3,500	17,500
Paying users	20	100	1,000	5,000
Premium subscribers	2	10	100	750
Overall funnel conversion (awareness to paying)	4.0%	2.0%	2.0%	2.0%

Registration to first consult	25%	25%	25%	30%
MAU to paying	20%	29%	29%	29%

6.3 Admin Ops Impact on Funnel Conversion

Admin ops staff improve conversion rates at two critical funnel stages:

Funnel Stage	Without Admin Ops	With Admin Ops	Lift	Mechanism
Download to Registration	40%	55%	+15pp	Admin ops calls back users who started but did not complete registration
Registration to First Consultation	15%	25%	+10pp	Admin ops proactively books first consultation for new registrants within 48 hours
Lapsed user re-engagement	0% (no mechanism)	5-10% reactivation	+5-10pp	Admin ops calls users who registered 30+ days ago with no activity

6.4 Funnel Optimization Priorities by Phase

Phase	Bottleneck	Optimization	Expected Lift	Owner
Pre-Pilot	Registration to first use	Onboarding tutorial, demo consultation, admin ops test calls	+10-15% activation	Product + Admin Ops
Pilot 1	Awareness (top of funnel)	Facility-driven distribution, WhatsApp, admin ops callbacks	3-5x awareness reach	Founders + Admin Ops
Pilot 2	MAU to paying	Push notifications, appointment reminders, consult quality, admin ops follow-up	+5-10% conversion	Product + Admin Ops
Production	Download to registration	Simplified signup, mobile-money-first flow, admin ops assisted registration	+10-20% registration rate	Product + Admin Ops

7. Scenario Analysis — All 21 Combinations

7.1 Scenario 1: Fully Self-Funded (S1)

No external investor at any point. All marketing spend comes from founders' monthly self-funding budget. Growth is constrained by the portion of monthly budget allocable to acquisition after development, infrastructure, and operational staffing costs.

S1-L1 (Bootstrapped — ~\$500/mo total budget)

Marketing budget allocation: \$0/mo Pre-Pilot through Pilot 1. ~\$50/mo available from Pilot 2 onward (10% of total budget after dev + infra + ops).

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-8)	6 mo	\$0/mo	4	400	\$0	\$0	\$0	N/A
Pilot 2 (M8-18)	10 mo	\$50/mo (\$500 total)	6	2,000	\$100	\$0.20	\$0.31	48.1x
Production (M18+)	Ongoing	\$100/mo	8	5,000	\$150	\$0.50	\$0.60	24.8x

Channels: 100% organic and facility-driven through Pilot 1. Pilot 2: WhatsApp (\$0), admin ops callbacks (salaried), \$50/mo Facebook boosts. Production: \$60/mo Facebook, \$40/mo SMS. Admin ops staff handle inquiry conversion at all phases.

Constraints: No paid advertising budget through Pilot 1. Growth entirely dependent on facility onboarding rate, admin ops callbacks, and organic word-of-mouth. Timeline to 5,000 users: 18-24 months.

S1-L2 (Ideal — ~\$1,000/mo total budget)

Marketing budget allocation: \$0/mo Pre-Pilot. \$100/mo from Pilot 1. \$200/mo from Pilot 2. \$300/mo at Production.

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$100/mo (\$300)	5	700	\$60	\$0.15	\$0.43	34.7x
Pilot 2 (M5-10)	5 mo	\$200/mo (\$1,000)	10	4,000	\$200	\$0.30	\$0.40	37.3x
Production (M10+)	Ongoing	\$300/mo	20	12,000	\$250	\$0.75	\$0.85	17.5x

Channels: Pilot 1: Founder sales + \$100/mo WhatsApp/social + admin ops callbacks. Pilot 2: Facebook (\$100/mo), Play Store ASO (\$50/mo), referral credits (\$50/mo), admin ops expanded to 4 staff. Production: Facebook/Instagram (\$150/mo), SMS (\$50/mo), referral (\$50/mo), ASO (\$50/mo).

S1-L3 (Fully Funded — ~\$2,000/mo total budget)

Marketing budget allocation: \$0/mo Pre-Pilot. \$200/mo from Pilot 1. \$500/mo from Pilot 2. \$700/mo at Production.

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	3	200	\$0	\$0	\$0	N/A

Pilot 1 (M2-5)	3 mo	\$200/mo (\$600)	5	900	\$120	\$0.25	\$0.67	22.2x
Pilot 2 (M5-10)	5 mo	\$500/mo (\$2,500)	15	7,000	\$200	\$0.40	\$0.44	33.9x
Production (M10+)	Ongoing	\$700/mo	35	20,000	\$200	\$1.00	\$1.05	14.2x

Channels: Pilot 1: Founder sales + WhatsApp + social (\$200/mo) + admin ops. Pilot 2: Facebook (\$200/mo), ASO (\$100/mo), referral (\$100/mo), industry event (\$500 one-time). Production: Full digital mix (\$400/mo) + referral (\$150/mo) + SMS (\$100/mo) + quarterly events (\$500).

7.2 Scenario 2: Investor Arrives at Pilot 2 Transition (S2)

Self-funded from Month 0 through Pilot 1. Investor capital arrives before or during Pilot 2, unlocking marketing budget for scaled acquisition. Pre-investor phases use the same acquisition approach as the corresponding S1 level.

S2-O1-I1 (Bootstrapped Self + Bootstrapped Investor)

Pre-investor: Identical to S1-L1 through Pilot 1. **Post-investor marketing allocation:** 20-25% of \$25-40K = \$5,000-10,000 over 12-18 months (~\$400-700/mo).

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-8)	6 mo	\$0/mo	4	400	\$0	\$0	\$0	N/A
Pilot 2 (M8-14)	6 mo	\$400/mo (\$2,400)	10	4,000	\$250	\$0.60	\$0.72	20.7x
Production (M14+)	Ongoing	\$500/mo	18	10,000	\$250	\$1.00	\$1.10	13.5x

Total marketing spend to Production: ~\$8,400

S2-O1-I2 (Bootstrapped Self + Ideal Investor)

Pre-investor: Identical to S1-L1 through Pilot 1. **Post-investor marketing allocation:** 20-25% of \$75-100K = \$15,000-25,000 (~\$1,000-1,500/mo).

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-8)	6 mo	\$0/mo	4	400	\$0	\$0	\$0	N/A
Pilot 2 (M8-14)	6 mo	\$1,200/mo (\$7,200)	15	7,000	\$200	\$1.00	\$1.04	14.3x
Production (M14+)	Ongoing	\$1,500/mo	30	20,000	\$200	\$1.50	\$1.55	9.6x

Total marketing spend to Production: ~\$22,200

S2-O1-I3 (Bootstrapped Self + Full Investor)

Pre-investor: Identical to S1-L1 through Pilot 1. **Post-investor marketing allocation:** 20-25% of \$150-250K = \$30,000-62,500 (~\$2,000-3,500/mo).

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-8)	6 mo	\$0/mo	4	400	\$0	\$0	\$0	N/A
Pilot 2 (M8-14)	6 mo	\$2,500/mo (\$15,000)	20	10,000	\$250	\$1.50	\$1.53	9.7x
Production (M14+)	Ongoing	\$3,500/mo	50	35,000	\$200	\$2.00	\$2.03	7.3x

Total marketing spend to Production: ~\$57,000

S2-O2-I1 (Ideal Self + Bootstrapped Investor)

Pre-investor: Identical to S1-L2 through Pilot 1. **Post-investor marketing allocation:** \$5,000-10,000 total (~\$400-700/mo), supplementing ongoing \$200/mo from self-funding.

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$100/mo (\$300)	5	700	\$60	\$0.15	\$0.43	34.7x
Pilot 2 (M5-10)	5 mo	\$600/mo (\$3,000)	12	5,000	\$250	\$0.60	\$0.66	22.6x
Production (M10+)	Ongoing	\$700/mo	22	15,000	\$250	\$1.00	\$1.05	14.2x

Total marketing spend to Production: ~\$10,300

S2-O2-I2 (Baseline Plan — Ideal Self + Ideal Investor)

This is the baseline plan for Health Hub. Pre-investor: S1-L2 pacing. Post-investor: marketing budget of \$15,000-25,000 (~\$1,200-1,500/mo).

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A

Pilot 1 (M2-5)	3 mo	\$100/mo (\$300)	5	700	\$60	\$0.15	\$0.43	34.7x
Pilot 2 (M5-10)	5 mo	\$1,500/mo (\$7,500)	15	8,000	\$200	\$1.00	\$1.02	14.6x
Production (M10+)	Ongoing	\$2,000/mo	35	25,000	\$200	\$1.50	\$1.53	9.7x

Total marketing spend to Production: ~\$23,800

Channel mix at Production (S2-O2-I2):

Channel	Monthly Budget	Share	Est. Users/Mo
Facebook / Instagram paid	\$600	30%	150-200
Google Play Store (ASO + paid)	\$300	15%	150-300
Referral program credits	\$300	15%	150-200
SMS campaigns	\$200	10%	200-400
Admin ops callbacks (salaried, allocated)	\$0 incremental	Embedded	80-120
Industry events (amortized)	\$200	10%	10-20 (B2B leads)
B2B sales travel	\$150	7.5%	1-2 facilities
Content marketing	\$150	7.5%	50-100
WhatsApp promotions	\$100	5%	50-100

S2-O2-I3 (Ideal Self + Full Investor)

Pre-investor: S1-L2 pacing. **Post-investor:** \$30,000-62,500 marketing budget (~\$2,500-3,500/mo).

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$100/mo (\$300)	5	700	\$60	\$0.15	\$0.43	34.7x
Pilot 2 (M5-10)	5 mo	\$3,000/mo (\$15,000)	20	10,000	\$250	\$1.50	\$1.53	9.7x
Production (M10+)	Ongoing	\$4,000/mo	60	40,000	\$200	\$2.50	\$2.50	6.0x

Total marketing spend to Production: ~\$63,300

S2-O3-I1 (Full Self + Bootstrapped Investor)

Pre-investor: Identical to S1-L3 through Pilot 1. The \$25-40K investor injection adds limited incremental marketing on top of already-robust L3 self-funding.

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	3	200	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$200/mo (\$600)	5	900	\$120	\$0.25	\$0.67	22.2x
Pilot 2 (M5-10)	5 mo	\$700/mo (\$3,500)	15	7,500	\$200	\$0.50	\$0.55	27.1x
Production (M10+)	Ongoing	\$900/mo	30	22,000	\$200	\$1.00	\$1.05	14.2x

Total marketing spend to Production: ~\$12,100

S2-O3-I2 (Full Self + Ideal Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	3	200	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$200/mo (\$600)	5	900	\$120	\$0.25	\$0.67	22.2x
Pilot 2 (M5-10)	5 mo	\$2,000/mo (\$10,000)	20	10,000	\$250	\$1.00	\$1.06	14.1x
Production (M10+)	Ongoing	\$2,500/mo	45	30,000	\$200	\$1.50	\$1.53	9.7x

Total marketing spend to Production: ~\$35,600

S2-O3-I3 (Full Self + Full Investor — Maximum S2)

Maximum Scenario 2 configuration. Robust self-funded base plus full investor marketing budget.

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	3	200	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$200/mo (\$600)	5	900	\$120	\$0.25	\$0.67	22.2x
Pilot 2 (M5-10)	5 mo	\$4,000/mo (\$20,000)	25	12,000	\$250	\$1.70	\$1.72	8.7x

Production (M10+)	Ongoing	\$5,000/mo	75	50,000	\$200	\$2.50	\$2.50	6.0x
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Total marketing spend to Production: ~\$81,600

7.3 Scenario 3: Investor Arrives Right After Pilot 1 (S3)

Self-funded only from Month 0 through early Pilot 1 (0-3 months self-funded runway). Investor capital arrives immediately after Pilot 1, funding the entire Pilot 2 and Production journey. This is the fastest path to scaled acquisition.

S3-O1-I1 (Bootstrapped Bridge + Bootstrapped Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$0/mo	3	300	\$0	\$0	\$0	N/A
Pilot 2 (M5-10)	5 mo	\$500/mo (\$2,500)	8	4,000	\$300	\$0.60	\$0.69	21.6x
Production (M10+)	Ongoing	\$600/mo	15	10,000	\$250	\$1.00	\$1.05	14.2x

Total marketing spend to Production: ~\$8,500

S3-O1-I2 (Bootstrapped Bridge + Ideal Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$0/mo	3	300	\$0	\$0	\$0	N/A
Pilot 2 (M5-10)	5 mo	\$1,500/mo (\$7,500)	15	8,000	\$200	\$1.00	\$1.00	14.9x
Production (M10+)	Ongoing	\$2,000/mo	35	25,000	\$200	\$1.50	\$1.53	9.7x

Total marketing spend to Production: ~\$25,500

S3-O1-I3 (Bootstrapped Bridge + Full Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$0/mo	3	300	\$0	\$0	\$0	N/A
Pilot 2 (M5-10)	5 mo	\$3,000/mo (\$15,000)	20	10,000	\$250	\$1.50	\$1.53	9.7x
Production (M10+)	Ongoing	\$4,000/mo	60	40,000	\$200	\$2.50	\$2.50	6.0x

Total marketing spend to Production: ~\$63,000

S3-O2-I1 (Ideal Bridge + Bootstrapped Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$100/mo (\$300)	5	600	\$60	\$0.15	\$0.50	29.8x
Pilot 2 (M5-10)	5 mo	\$600/mo (\$3,000)	12	5,000	\$250	\$0.60	\$0.66	22.6x
Production (M10+)	Ongoing	\$700/mo	22	15,000	\$250	\$1.00	\$1.05	14.2x

Total marketing spend to Production: ~\$10,300

S3-O2-I2 (Ideal Bridge + Ideal Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$100/mo (\$300)	5	600	\$60	\$0.15	\$0.50	29.8x
Pilot 2 (M5-10)	5 mo	\$1,500/mo (\$7,500)	18	9,000	\$200	\$1.00	\$1.03	14.5x
Production (M10+)	Ongoing	\$2,500/mo	40	30,000	\$200	\$1.50	\$1.53	9.7x

Total marketing spend to Production: ~\$27,800

S3-O2-I3 (Ideal Bridge + Full Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	2	150	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$100/mo (\$300)	5	600	\$60	\$0.15	\$0.50	29.8x

Pilot 2 (M5-10)	5 mo	\$3,500/mo (\$17,500)	25	12,000	\$250	\$1.50	\$1.52	9.8x
Production (M10+)	Ongoing	\$5,000/mo	70	50,000	\$200	\$2.50	\$2.50	6.0x

Total marketing spend to Production: ~\$73,800

S3-O3-I1 (Full Bridge + Bootstrapped Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	3	200	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$200/mo (\$600)	5	800	\$120	\$0.25	\$0.75	19.9x
Pilot 2 (M5-10)	5 mo	\$700/mo (\$3,500)	15	7,000	\$200	\$0.50	\$0.55	27.1x
Production (M10+)	Ongoing	\$900/mo	28	20,000	\$200	\$1.00	\$1.05	14.2x

Total marketing spend to Production: ~\$12,100

S3-O3-I2 (Full Bridge + Ideal Investor)

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	3	200	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$200/mo (\$600)	5	800	\$120	\$0.25	\$0.75	19.9x
Pilot 2 (M5-10)	5 mo	\$2,500/mo (\$12,500)	22	11,000	\$250	\$1.20	\$1.23	12.1x
Production (M10+)	Ongoing	\$3,000/mo	50	35,000	\$200	\$1.50	\$1.53	9.7x

Total marketing spend to Production: ~\$41,100

S3-O3-I3 (Maximum Configuration — Full Bridge + Full Investor)

This is the theoretical maximum configuration. Full self-funded bridge with the largest investor injection. Fastest path to scale.

Phase	Duration	Marketing Budget	Facilities (Cumul.)	Patients (Cumul.)	B2B CAC	B2C CAC	Blended CAC	LTV:CAC
Pre-Pilot (M0-2)	2 mo	\$0/mo	3	200	\$0	\$0	\$0	N/A
Pilot 1 (M2-5)	3 mo	\$200/mo (\$600)	5	800	\$120	\$0.25	\$0.75	19.9x
Pilot 2 (M5-10)	5 mo	\$5,000/mo (\$25,000)	30	15,000	\$250	\$1.70	\$1.72	8.7x
Production (M10+)	Ongoing	\$6,000/mo	100	60,000	\$200	\$3.00	\$3.00	5.0x

Total marketing spend to Production: ~\$97,600

Channel mix at Production (S3-O3-I3):

Channel	Monthly Budget	Share	Est. Users/Mo
Facebook / Instagram paid	\$1,500	25%	300-500
Google Ads (Play Store + Search)	\$900	15%	300-600
Referral program credits	\$600	10%	300-400
SMS / Telecom campaigns	\$600	10%	600-1,200
Radio advertising	\$500	8%	200-400 (est.)
B2B sales team (part-time)	\$500	8%	3-5 facilities
Content marketing (video, blog)	\$400	7%	100-200
Industry events (amortized)	\$400	7%	2-3 facilities
Community health worker incentives	\$300	5%	100-200
WhatsApp promotions	\$200	3%	100-200
Admin ops callbacks (salaried)	\$0 incremental	Embedded	200-400
Reserve / testing	\$100	2%	Experimental

7.4 Summary Table — All 21 Combinations

#	Combo	Our Level	Investor Level	Total Mktg Spend to Prod.	Facilities at Prod.	Users at Prod.	Blended CAC	LTV:CAC	Time to 10K Users
1	S1-L1	Bootstrapped	None	\$1,400	8	5,000	\$0.60	24.8x	24+ mo
2	S1-L2	Ideal	None	\$3,900	20	12,000	\$0.85	17.5x	14 mo
3	S1-L3	Fully Funded	None	\$9,100	35	20,000	\$1.05	14.2x	10 mo

4	S2-O1-I1	Bootstrapped	Bootstrapped	\$8,400	18	10,000	\$1.10	13.5x	14 mo
5	S2-O1-I2	Bootstrapped	Ideal	\$22,200	30	20,000	\$1.55	9.6x	12 mo
6	S2-O1-I3	Bootstrapped	Fully Funded	\$57,000	50	35,000	\$2.03	7.3x	10 mo
7	S2-O2-I1	Ideal	Bootstrapped	\$10,300	22	15,000	\$1.05	14.2x	12 mo
8	S2-O2-I2	Ideal	Ideal	\$23,800	35	25,000	\$1.53	9.7x	10 mo
9	S2-O2-I3	Ideal	Fully Funded	\$63,300	60	40,000	\$2.50	6.0x	8 mo
10	S2-O3-I1	Fully Funded	Bootstrapped	\$12,100	30	22,000	\$1.05	14.2x	10 mo
11	S2-O3-I2	Fully Funded	Ideal	\$35,600	45	30,000	\$1.53	9.7x	9 mo
12	S2-O3-I3	Fully Funded	Fully Funded	\$81,600	75	50,000	\$2.50	6.0x	7 mo
13	S3-O1-I1	Bootstrapped	Bootstrapped	\$8,500	15	10,000	\$1.05	14.2x	12 mo
14	S3-O1-I2	Bootstrapped	Ideal	\$25,500	35	25,000	\$1.53	9.7x	10 mo
15	S3-O1-I3	Bootstrapped	Fully Funded	\$63,000	60	40,000	\$2.50	6.0x	8 mo
16	S3-O2-I1	Ideal	Bootstrapped	\$10,300	22	15,000	\$1.05	14.2x	12 mo
17	S3-O2-I2	Ideal	Ideal	\$27,800	40	30,000	\$1.53	9.7x	9 mo
18	S3-O2-I3	Ideal	Fully Funded	\$73,800	70	50,000	\$2.50	6.0x	7 mo
19	S3-O3-I1	Fully Funded	Bootstrapped	\$12,100	28	20,000	\$1.05	14.2x	10 mo
20	S3-O3-I2	Fully Funded	Ideal	\$41,100	50	35,000	\$1.53	9.7x	8 mo
21	S3-O3-I3	Fully Funded	Fully Funded	\$97,600	100	60,000	\$3.00	5.0x	6 mo

7.5 Efficiency Analysis

Most capital-efficient combinations (highest users per dollar of marketing spend):

Rank	Combo	Users per \$1 Marketing	Notes
1	S1-L1	3.57 users/\$1	Almost entirely organic + admin ops; time-intensive but capital-efficient
2	S1-L2	3.08 users/\$1	Good balance of spend and organic leverage
3	S1-L3	2.20 users/\$1	Self-funded efficiency before paid channels dominate
4	S2-O2-I1 / S3-O2-I1	1.46 users/\$1	Modest investor keeps CAC low
5	S2-O3-I1 / S3-O3-I1	1.65-1.82 users/\$1	Robust self-fund base with minimal investor increment

Fastest to 10,000 users:

Rank	Combo(s)	Time to 10K Users	Total Marketing Spend
1	S3-O3-I3	6 months	\$97,600
2	S3-O2-I3, S2-O3-I3	7 months	\$73,800 / \$81,600
3	S3-O1-I3, S2-O2-I3, S3-O3-I2	8 months	\$63,000 / \$63,300 / \$41,100
4	S3-O2-I2, S2-O3-I2	9 months	\$27,800 / \$35,600
5	S2-O2-I2 (Baseline), S3-O1-I2, S1-L3, S2-O3-I1	10 months	\$23,800 / \$25,500 / \$9,100 / \$12,100

7.6 LTV:CAC Risk Zones

LTV:CAC Range	Verdict	Combos in This Range
14x+	Excellent (organic-dominated, high efficiency)	S1-L1, S1-L2, S1-L3, S2-O2-I1, S2-O3-I1, S3-O1-I1, S3-O2-I1, S3-O3-I1, S2-O1-I2
9x-14x	Very Good (balanced paid + organic)	S2-O2-I2, S2-O3-I2, S3-O1-I2, S3-O2-I2, S3-O3-I2, S2-O1-I3
5x-9x	Good (paid channels scaling)	S2-O2-I3, S2-O3-I3, S3-O1-I3, S3-O2-I3, S2-O1-I1, S3-O3-I3
3x-5x	Acceptable (approaching efficiency floor)	None in current model
Below 3x	Warning (CAC exceeds sustainable threshold)	None

Key insight: No combination falls below the 5x LTV:CAC threshold. Even the maximum-spend scenario (S3-O3-I3) maintains a 5.0x ratio because (a) East African digital advertising is 3-5x cheaper than Western markets, (b) facility- driven acquisition provides a zero-cost base layer, and (c) admin ops callbacks add a low-CAC manual conversion channel that buffers blended CAC even as paid channels scale.

8. Retention and Churn Strategy

8.1 Monthly Churn Targets

Metric	Pilot 1	Pilot 2	Production (Early)	Production (Mature)	Industry Benchmark
Monthly churn (overall)	15%	10%	7%	5%	5-10% (health SaaS)
Monthly churn (premium subscribers)	10%	7%	5%	3%	3-5% (subscription)
D1 retention (day after registration)	40%	50%	55%	60%	25-35% (health apps)
D7 retention	25%	30%	35%	40%	15-20%
D30 retention	15%	20%	25%	30%	8-12%
NPS (Net Promoter Score)	20	35	45	50+	30-40 (health tech)

8.2 Retention Tactics by Phase

Strategy	Phase	Implementation Cost	Expected Impact	Dependencies
Onboarding tutorial (in-app walkthrough)	Pilot 1+	\$0 (dev time)	+15% activation rate	Android/web app
Admin ops onboarding callback (within 48 hrs of registration)	Pilot 1+	\$0 incremental (salaried staff)	+10% first-use conversion	Admin ops staff
Push notifications for appointments	Pilot 1+	\$0 (Firebase free tier)	+10% 30-day retention	Android app
WhatsApp appointment reminders	Pilot 1+	\$0.02-0.05/message	+20% return visit rate	WhatsApp Business API
Email health tips and summaries	Pilot 1+	\$0 (SendGrid free tier)	+5% monthly retention	Email service
Admin ops lapsed-user callbacks	Pilot 2+	\$0 incremental (salaried staff)	+5-10% reactivation of 30-day lapsed users	Admin ops staff
Referral rewards (give ETB 50, get ETB 50)	Pilot 2+	\$1.75/successful referral	+10% referrer retention	Payment system
Premium subscription benefits	Pilot 2+	\$0 (feature gating)	-30% churn vs. free users	Subscription billing
Gamification (health score, streaks)	Pilot 2+	\$0 (dev time)	+15% DAU/MAU ratio	Frontend dev
Loyalty program (points per consultation)	Production	\$0.10-0.25/point redeemed	-20% overall churn	Points system
Family/dependent accounts	Production	\$0 (dev time)	+30% household retention	Dependent profile feature
Offline mode (cached health records)	Production	\$0 (dev time)	+10% retention in low-connectivity	Service worker

8.3 Churn Analysis

Churn Trigger	Frequency	Mitigation	Measurement
Poor first experience (long wait, tech issues)	High in Pilot 1	Onboarding tutorial, test consultations, quality monitoring, admin ops quality callback	First-session drop-off rate
No perceived need (healthy user)	Ongoing	Health tips, preventive care reminders, seasonal content	Days since last activity
Price sensitivity	High for students, rural	Freemium tier, consultation discounts, subscription value	Free-to-paid conversion rate
Provider quality concerns	Medium	Provider ratings, quality metrics, feedback loops, admin ops follow-up	Post-consultation NPS
Competitor offering	Low (early market)	Feature velocity, network effects, switching costs	Competitive win/loss
App performance / UX issues	Medium	Performance monitoring, crash reporting, iterative UX	App store rating, crash rate
Admin ops unavailability (callbacks missed)	Medium in Pilot 1	Expand admin ops staffing per params.md ladder, SLA on callback time	Callback response time, missed callback rate

8.4 Retention Economics

Improvement	Impact on LTV	Equivalent Marketing Spend
Reduce monthly churn from 15% to 10%	+50% customer lifespan (6.7 mo to 10 mo)	Equivalent to \$5,000-10,000 in new user acquisition
Reduce monthly churn from 10% to 7%	+43% customer lifespan (10 mo to 14.3 mo)	Equivalent to \$8,000-15,000 in new user acquisition
Reduce monthly churn from 7% to 5%	+40% customer lifespan (14.3 mo to 20 mo)	Equivalent to \$10,000-25,000 in new user acquisition
Increase ARPU from \$1.35 to \$1.80	+33% LTV	Requires ~4% more users on premium subscription

9. Key Metrics Dashboard

9.1 Core Metrics by Phase (Baseline Plan: S2-O2-I2)

Metric	Pre-Pilot	Pilot 1	Pilot 2	Production
Total registered users	200	1,000	10,000	50,000

Monthly active users (MAU)	100	350	3,500	17,500
Daily active users (DAU)	20	70	700	5,000
DAU/MAU ratio	20%	20%	20%	29%
B2B facilities onboarded	3	5	15	35
Patients per facility	67	200	667	1,429
Blended CAC	\$0	\$0.43	\$1.02	\$1.53
ARPU (monthly)	\$0	\$0.50	\$1.00	\$1.50
LTV (moderate)	N/A	\$6.00	\$12.00	\$18.00
LTV:CAC	N/A	14.0x	11.8x	11.8x
Monthly churn (overall)	N/A	15%	10%	7%
Monthly churn (premium)	N/A	10%	7%	5%
Viral coefficient	0	0.1	0.3	0.5
Payback period (months)	N/A	<1	1	1-2
Admin ops conversions/mo	0	15-30	80-160	250-500

9.2 CAC, LTV, and Payback by Phase

Phase	Blended CAC	LTV (Moderate)	LTV:CAC	Payback Period	Notes
Pre-Pilot	\$0	N/A	N/A	N/A	Internal users only
Pilot 1	\$0.43	\$6.00	14.0x	<1 month	Organic/facility-driven dominates
Pilot 2	\$1.02	\$12.00	11.8x	~1 month	Paid channels entering mix
Production	\$1.53	\$18.00	11.8x	1-2 months	Paid channels at scale

9.3 Channel Performance Tracking

Channel	Primary Metric	Secondary Metric	Tracking Method
Facility-driven	Registrations per facility/mo	Activation rate of facility-referred users	UTM on facility-specific signup links
Admin ops callbacks	Callback-to-registration conversion rate	Time-to-first-consultation for admin-acquired users	CRM / call center software tracking
WhatsApp	Click-through rate on shared links	Registrations from WhatsApp UTM	UTM tracking, WhatsApp Business analytics
Social media organic	Engagement rate, link clicks	Cost per registration (time-adjusted)	Meta Business Suite
Social media paid	CPC, CPI (cost per install)	ROAS (return on ad spend)	Meta Ads Manager
Play Store	Organic installs, keyword ranking	Install-to-registration conversion	Google Play Console, Firebase
Referral program	Referrals sent/user, conversion rate	Viral coefficient	In-app referral tracking
SMS	Delivery rate, CTR	Cost per registration	SMS gateway analytics

9.4 North Star Metrics

Phase	North Star Metric	Target	Rationale
Pre-Pilot	Completed test consultations	50	Validates core flow works end-to-end
Pilot 1	Facilities with 100+ registered patients	3	Proves facility-driven acquisition model
Pilot 2	Paying users (at least 1 paid consultation)	1,000	Validates willingness to pay at scale
Production	Monthly revenue	\$5,000+	Path to success metric (revenue >= investment)

9.5 Viral Coefficient Progression

Phase	Viral Coefficient	Meaning	Driver
Pre-Pilot	0.0	No organic spread	Internal testing only
Pilot 1	0.1	Each 10 users generate 1 new user	Word of mouth from satisfied pilot patients
Pilot 2	0.3	Each 10 users generate 3 new users	Referral program + WhatsApp sharing + admin ops-prompted referrals
Production	0.5	Each 10 users generate 5 new users	Mature referral program + social proof + network effects

Note: A viral coefficient of 1.0 would mean self-sustaining growth. Health apps rarely achieve this. The 0.5 target at Production is ambitious but achievable with strong referral incentives and admin ops prompting satisfied patients to refer friends/family.

10. Key Takeaways

1. **Facility-first acquisition is the highest-leverage strategy.** Each onboarded healthcare facility generates 50-200 patient registrations at near-zero marginal cost. Through Pilot 2, facility-driven acquisition accounts for 30-50% of all users across every combination. The B2B sales motion — founder-led, then referral-driven, then admin ops-supported — is the single most important go-to-market activity.
2. **Admin ops staff are a unique acquisition channel.** Unlike pure-play tech platforms, Health Hub employs admin ops staff who handle callbacks, onboarding assistance, and inquiry conversion. This manual channel delivers users at \$0.30-1.00 effective CAC and improves funnel conversion at the two most critical stages (registration and first consultation). No competitor in the East African health tech space operates this model.
3. **All 21 combinations achieve viable unit economics.** LTV:CAC ratios range from 5.0x (S3-O3-I3, maximum spend) to 24.8x (S1-L1, maximum efficiency). No combination enters the sub-3x danger zone because East African digital advertising costs are 3-5x cheaper than Western markets, facility-driven acquisition provides a zero-cost base layer, and admin ops callbacks add a low-CAC manual conversion channel.
4. **The baseline plan (S2-O2-I2) balances speed and efficiency.** At \$23,800 total marketing spend to Production, the baseline achieves 25,000 users and 35 facilities with a 9.7x LTV:CAC ratio. This is the recommended acquisition plan for investor conversations: it demonstrates both capital discipline and a credible path to scale.
5. **WhatsApp and Play Store ASO are the two highest-ROI scaled channels.** WhatsApp communities cost \$0-0.25 per acquired user and leverage existing social networks. Play Store ASO (post-Android launch) costs \$0.50-2.00 per install and scales with app store ratings. Both should receive disproportionate attention relative to paid social advertising.
6. **Retention is the multiplier.** A 5-percentage-point improvement in monthly churn (e.g., from 10% to 5%) doubles average customer lifespan and doubles LTV. Investment in push notifications, WhatsApp reminders, admin ops follow-up calls, and premium subscription value delivers higher ROI than equivalent spending on top-of-funnel acquisition.
7. **Payback period is under 2 months in all phases.** Because blended CAC stays in the \$0-3 range and monthly ARPU reaches \$1-1.50 at scale, the time to recover acquisition cost from a single user is consistently short. This allows aggressive reinvestment of early revenue into growth.
8. **The admin ops acquisition channel scales with staffing, not with budget.** Unlike paid digital channels where CAC rises with competition, admin ops conversion capacity scales linearly with headcount at a fixed per-staff cost (\$217/mo per per params.md). This creates a predictable, budget-insulated acquisition pathway.

Cross-References

Document	Relevance to Customer Acquisition
params.md	Scenario definitions, investment levels, phase timelines, market parameters, FX rates, admin ops staffing and salary benchmarks
06-pricing-strategy	Consultation pricing (ETB 200 GP, ETB 400 Specialist), subscription pricing (ETB 300/mo), impact on ARPU and LTV
05-revenue-modelling	Revenue projections by phase dependent on user acquisition targets modeled here
04-effort-estimation	Development effort for acquisition-enabling features (referral system, push notifications, ASO, WhatsApp integration)
02-services-slas	Service quality commitments that directly impact retention and churn rates

Changelog

Date	Version	Change
2026-03	2.0	v2 rewrite: added admin ops as acquisition channel, updated LTV model with care coordination revenue, revised all 21 scenario tables with v2 params, added funnel impact analysis for admin ops, expanded retention economics

End of 08-customer-acquisition.md (v2)